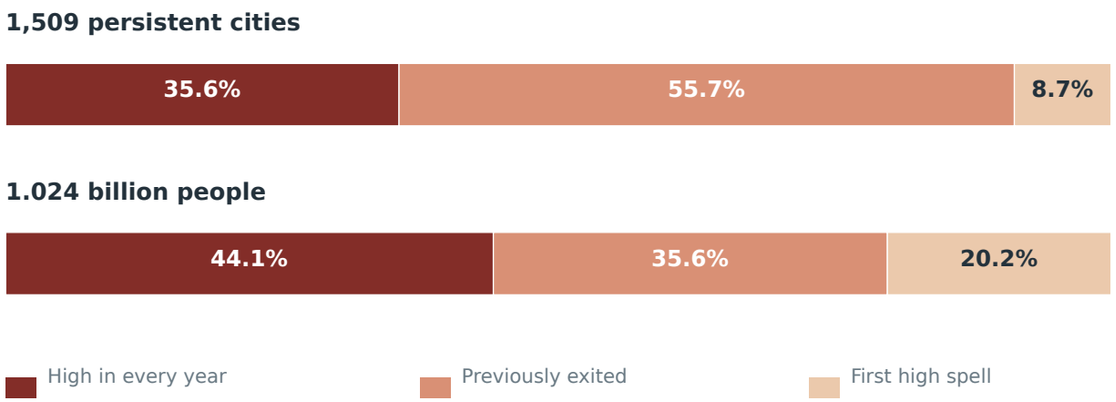
GLOBAL URBAN HEAT

Persistent-city accumulation reflects turnover and duration-dependent exit

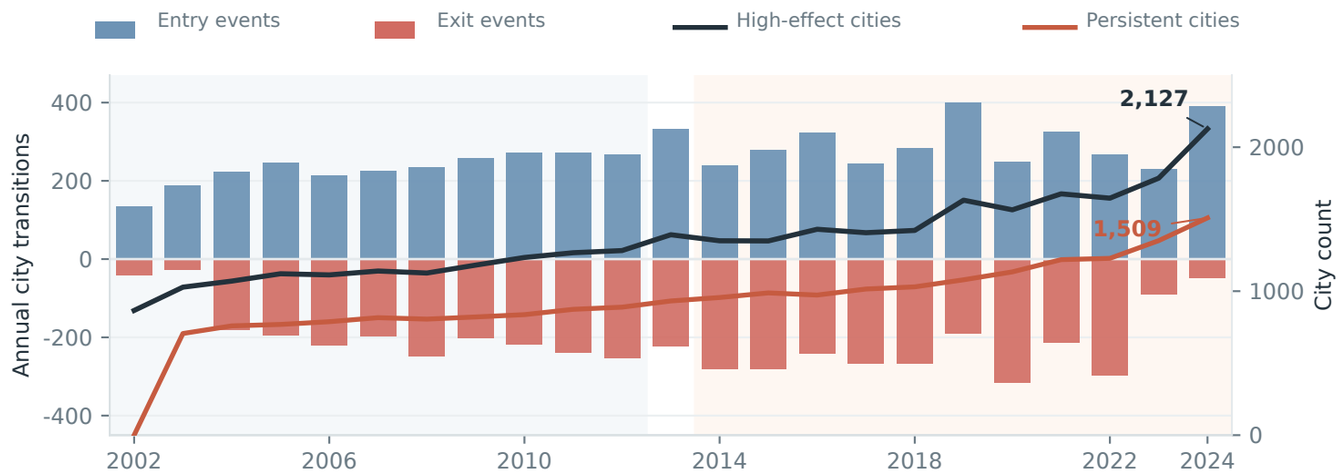
23 annual transitions / 6,081 high-state entries / 4,726 exits

A Historical composition of persistent cities



537 throughout 2001–2024 · 840 with a previous exit · 132 in their first high spell

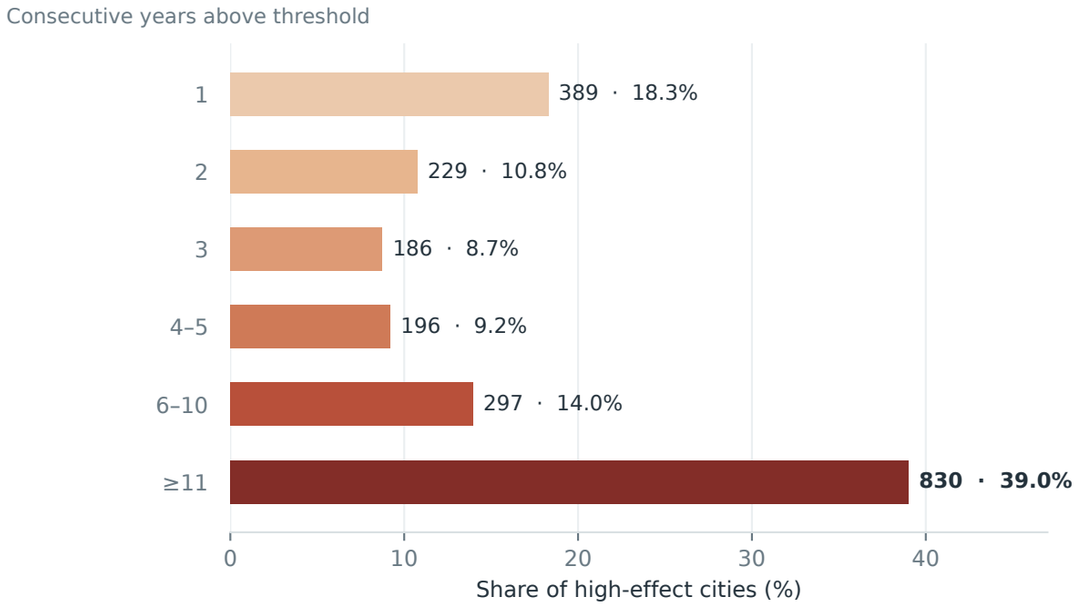
B Turnover persisted as the city pool expanded



Entry 2.66% · Exit 16.92%

Entry 3.57% · Exit 14.94%

C The duration structure in 2024



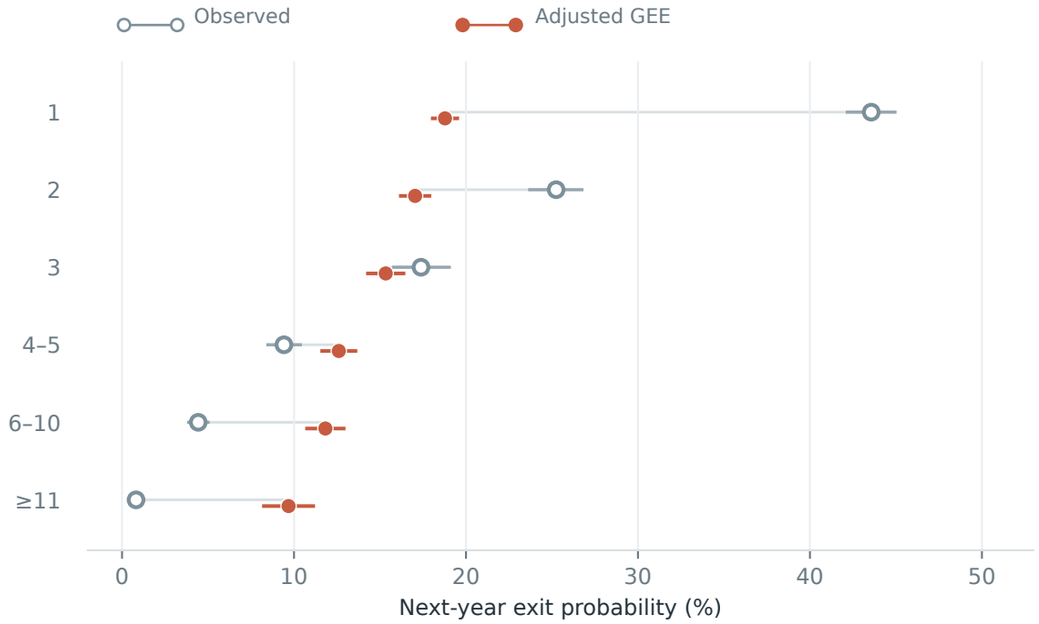
2,127

high-effect cities

39.0%

high for ≥11 years

D Longer duration is associated with lower exit



Adjusted OR per doubling 0.735

95% CI 0.702–0.770 · 2,721 cities · 29,853 at-risk city-years

A: 2024 persistent-city history. B: annual transitions. C: 2024 high-city durations. D: next-year exits; city-cluster bootstrap and GEE-adjusted intervals.